

12 (a) (i)	(gradual) loss of power/intensity/amplitude (not “signal”)	B1	[1]
(ii)	e.g. noise can be eliminated (not “there is no noise”) because pulses can be regenerated	M1 A1	
	e.g. much greater data handling/carrying capacity because many messages can be carried at the same time/greater bandwidth	M1 A1	
	e.g. more secure because it can be encrypted	(M1) (A1)	
	e.g. error checking because extra information/parity bit can be added	(M1) (A1)	[4]
	<i>(allow any two sensible suggestions with ‘state’ M1 and ‘explain’ A1)</i>		
(b)	attenuation = $10 \lg(145/29)$ (= 7.0)	C1	
	attenuation per unit length = $7.0/36$ = 0.19 dB km^{-1}	A1	[2]